

CRISPRa transplant screen

Millman lab | Maestas et al., 2026
Day-10 kidney graft endpoint versus differentiated day 0 (D0).

106,400

Published guides analyzed

20 / 20

Published top-20 genes recovered

59.94 sec

Total measured computation

MAGeCK results, with the evidence behind each candidate

SplicR combines MAGeCK RRA with sample-quality checks, guide-support flags and a traceable report. An independent MAGeCK run matched **all 13 numeric gene-result columns exactly**. SplicR also recovered the publication's **same top-20 genes**, with some ordering differences.

1. Was the guide library still represented?

Sample	Guides detected	Guides with zero counts	Mean counts per guide
Differentiated D0	106,358 / 106,400 99.96%	42	628.9
Day-10 kidney endpoint	106,358 / 106,400 99.96%	42	578.6

Only five guides detected at D0 were absent at the endpoint. These are aggregate counts, so individual-graft representation cannot be assessed.

2. Which ranked candidates have support across guides?

Gene	Enrichment rank	Guides moving upward	Largest guide share*
SIX3	1	5 of 5	27%
FCAMR	8	4 of 5	36%
GSTM3	10	4 of 5	28%
HSF2	2	2 of 5	89%

41 of the 190 top-ranked candidates received a guide-support flag. Each flag includes its measured evidence for review before follow-up experiments.

*Share of the sum of absolute guide log2 fold changes. Flags prompt review; they do not establish false positives.

Candidate ranking is separate from significance. No gene passed 10% directional FDR in either normalization run. These are exploratory candidates, and this reanalysis does not replace the paper's functional validation.

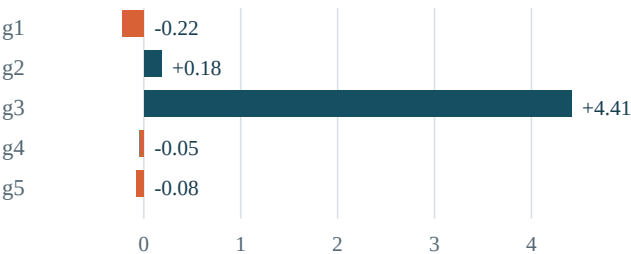
Evidence for follow-up

A concrete example from your data: FCAMR versus HSF2.

FCAMR



HSF2



Per-guide log2 fold changes on the same scale. Positive values mean increased endpoint representation; orange bars mean decreased representation.

FCAMR: four guides move upward; no single-guide flag.
HSF2: ranked second, but one guide contributes 89% of the absolute guide effect. SplicR flags it using its 60% concentration threshold.

3. Does the result depend on normalization?

Using the **3,755 non-targeting controls** for normalization, SIX3, FCAMR and GSTM3 kept positive gene effects and ranks **1, 9 and 11**, versus **1, 8 and 10** in the main run. The named candidates remain high-ranked under both choices.

How we produced the analysis

Step	What we did
Validate and QC	Checked all guide IDs, sequences and counts; confirmed D0 as the reference; measured guide detection, depth and count distribution.
Rank and review	Ran MAGeCK RRA and preserved its statistics; applied SplicR guide-concentration and agreement checks.
Verify and export	Checked standalone MAGeCK agreement and control normalization; exported full gene/guide tables, settings and file hashes.

Measured computation: main SplicR pipeline **21.93 sec**; with input preparation **25.36 sec**; all analysis, verification and CSV exports **59.94 sec**. The total sums timed execution phases; download, setup and document creation are excluded.

For your next screen: review guide support before committing to overexpression and transplant follow-up. Counts for each graft would also allow checks of agreement between grafts.

Scope: no mouse-level reproducibility, mapping rate or validation probability was inferred. Shared-sequence gene-label ambiguities remain in the full audit.

Source: Maestas et al., DOI 10.1093/stcltm/szag012, Source_Data.xlsx. MAGeCK 0.5.9.5; same counts, contrast and median normalization in the independent check. Full tables accompany this brief.